Josh Rauvola

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OBJECTIVE

As a current University of Chicago student with extensive hands-on experience in machine learning and computer vision, I've applied my skills in predictive modeling and data analysis across diverse roles, including software development and data science. With proven expertise in deploying TensorFlow models and optimizing data pipelines, I am ready to contribute to Tesla's CV stack, to drive advancements in factory efficiency, app functionality, and customer engagement.

EDUCATION

UNIVERSITY OF CHICAGO, Chicago, IL *Sept 2023 – Dec 2024*

- Master's in Applied Data Science with a focus in AI/Machine Learning
- Current GPA 3.95/4.0
- Relevant Course Work: Deep Learning, Computer Vision, Machine Learning, Predictive Analytics and Big Data

CALIFORNIA POLYTECHNIC STATE UNIVERSITY, San Luis Obispo, CA *Sept 2018 - June 2022*

- Dual Bachelor of Science: Computer Science (Focus in AI/Machine Learning) & Economics (Focus in Quantitative Analysis)
- Graduated in four years while working full time and receiving two degrees

PROFESSIONAL EXPERIENCE

AMAZE HEALTH, Denver, CO *(Comprehensive virtual healthcare service provider)*
Data Engineer, Jan 2023 – Sept 2023

- Supported the development of new patient learning center by setting up data pipelines and implementing data migration practices that allowed for the efficient transfer of terabytes worth of data in real time
- Engineered systematic enrollment and billing system to help drive efficiency gains and streamline customer onboarding to reduce lost billing income and increasing overall efficiency of process by 86%

CRITICAL INSIGHT, Seattle, WA *(Managed Detection and Response Cyber-Security company)*
Software Developer, June 2021 – October 2022

- Built and deployed new pages and features for the company's internal customer health portal to improve analytical functionality and information display
- Designed predictive models for internal portal to provide better respondent experience using machine learning algorithms in TensorFlow and scikit libraries
- Architected and developed Cassandra database system for cleaning and deployment of 10+ years of AWS S3 bucket data

PRIME TRUST, Las Vegas, NV *(Financial infrastructure fintech and digital asset company)*
Data Scientist, October 2021 – June 2022

- Implemented natural language processing algorithms into data pipeline in Airflow improving reporting efficiency by 60%
- Worked collaboratively with finance team and data scientists to assist in design and implementation of quantitative data analysis (time series forecasting) in customer pricing models using TensorFlow and Spark
- Provided consultation to tech and business leaders on how to implement best practices for effective and accurate data flow

EAST SEATTLE PARTNERS, Seattle, WA *(Leading multi-asset growth equity firm)*
Financial and Technology Analyst Intern, June 2020 – Aug 2020

- Engineered company wide database system to analyze 20 years of investment cash flow information to provide key performance measures such as HPR, IRR, and FMV rates
- Conducted market research and prepared reports for consideration by executives for growth equity investments

RESEARCH PAPERS

PREDICTING HEDONIC ADAPTATIONS IN THE LABOR FORCE MARKET

- Compiled and analyzed 20 years of unemployment and volatility index data to see if machine learning models could accurately predict recessionary situations such as the large negative effects of COVID-19 on the labor force market
- Implemented five machine learning models linear regression, multi-layer perceptron, recurrent neural network, LSTM, and prophet with the best model receiving an accuracy score of 92%

CORE LANGUAGES

- TensorFlow, Spark
- SQL, Cassandra
- Python, R, C
- Pytorch, Keras

PERSONAL INTERESTS

- University of Chicago Track and Field Athlete
- 4:17 mile time